

COMP0078: Graphs in Machine Learning (Graph-based Semi-supervised Learning)

Course Notes based on Lectures by Mark Herbster

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*Revision notation: items marked with $\star$ are specifically listed in the revision notes file (`revision25-26-V1.pdf`).
Non-examinable content (Kirchhoff's laws, Parts VIII-X on p -Laplacian) has been removed.*

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1 Part I: Motivation*

1.1 Semi-supervised Learning (SSL)

Definition 1.1 (Semi-supervised Learning*). Given a dataset $\mathcal{S} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_\ell, y_\ell), \mathbf{x}_{\ell+1}, \dots, \mathbf{x}_n\}$, where usually $\ell \ll n$, infer a function f such that $f(\mathbf{x}_i) \approx y_i$.

The hypothesis of SSL is that the structure of the input space may be exploited to improve learning.

- **Cluster hypothesis:** If points are in the same cluster, they are likely to be of the same class.
- **Manifold hypothesis:** The (high-dimensional) data lie on a low-dimensional manifold.

1.2 Inductive vs Transductive Learning

Definition 1.2 (Inductive SSL*). Given a training sample, an inductive learner learns a function $f : \mathcal{X} \rightarrow \mathcal{Y}$ with the aim that $f(\mathbf{x}) \approx y$ for **all** $\mathbf{x} \in \mathcal{X}$ (including unseen data).

Definition 1.3 (Transductive SSL*). Given a training sample, a transductive learner learns a function $f(\mathbf{x}) \approx y$ **only** on the unlabeled data $f : \mathcal{X}_U \rightarrow \mathcal{Y}$ (where $\mathcal{X}_U = \{\mathbf{x}_j\}_{j=\ell+1}^n$).

2 Part II: Graphs*

2.1 Building Graphs*

Graphs can be **intrinsic** (e.g., protein interaction networks) or **built** using a similarity metric $d(p, q)$.

1. **k -Nearest Neighbors (k -NN):** Vertex i connects to j if j is among the k nearest neighbors.
2. **ϵ -ball:** Connect i and j if $d(i, j) < \epsilon$.
3. **Tree-based:** Minimum Spanning Tree (MST) or Shortest Path Tree (SPT).
4. **Weighted graph:** Edge weights $w_{ij} = e^{-c d(i, j)}$ (Gaussian kernel).
5. **Combo:** Combination of the above.

3 Part III: The Graph Laplacian*

3.1 Definitions

Let $G = (V, E)$ be a graph with n vertices.

- W : Symmetric non-negatively weighted adjacency matrix ($w_{ij} \geq 0$).
- D : Diagonal degree matrix, $D_{ii} = d_i = \sum_{j \neq i} w_{ij}$.

Definition 3.1 (Graph Laplacian*). The graph Laplacian is defined as $L(G) := D - W$, where W is the (symmetric, non-negatively weighted) adjacency matrix and $D := \text{diag}(d_1, \dots, d_n)$ is the diagonal matrix of (weighted) degrees. The set of edges is denoted $E(G) := \{(i, j) : w_{ij} > 0, i < j\}$.

3.2 Quadratic Form and Properties

We define a semi-inner product and semi-norm associated with the graph:

Definition 3.2 (Laplacian Semi-inner Product and Semi-norm).

$$\langle \mathbf{u}, \mathbf{v} \rangle_G := \mathbf{u}^\top L \mathbf{v} \quad (1)$$

$$\|\mathbf{u}\|_G^2 := \langle \mathbf{u}, \mathbf{u} \rangle_G \quad (2)$$

Theorem 3.3 (Laplacian Quadratic Form*). *For any vector $\mathbf{u} \in \mathbb{R}^n$:*

$$\mathbf{u}^\top L \mathbf{u} = \sum_{(i,j) \in E(G)} w_{ij} (u_i - u_j)^2 \quad (3)$$

Proof. Using Definition 3.1:

$$\begin{aligned} \mathbf{u}^\top L \mathbf{u} &= \mathbf{u}^\top (D - W) \mathbf{u} \\ &= \sum_{i,j} D_{ij} u_i u_j - \sum_{i,j} w_{ij} u_i u_j \\ &= \sum_i d_i u_i^2 - \sum_{i,j} w_{ij} u_i u_j \\ &= \sum_i \left(\sum_{j \neq i} w_{ij} \right) u_i^2 - \sum_{i,j} w_{ij} u_i u_j \\ &= \frac{1}{2} \sum_{i,j} w_{ij} u_i^2 + \frac{1}{2} \sum_{i,j} w_{ij} u_j^2 - \sum_{i,j} w_{ij} u_i u_j \quad (\text{by symmetry of } W) \\ &= \frac{1}{2} \sum_{i,j} w_{ij} (u_i^2 - 2u_i u_j + u_j^2) \\ &= \sum_{i < j} w_{ij} (u_i - u_j)^2 \end{aligned}$$

□

Proposition 3.4 (Properties of the Laplacian*). 1. L is positive semi-definite (PSD), since $\mathbf{u}^\top L \mathbf{u} \geq 0$ for all $\mathbf{u}$ (by Theorem 3.3).

2. The constant vector $\mathbf{1}$ is an eigenvector with eigenvalue 0.

3. If the graph is connected, $\mathbf{1}$ is the **only** eigenvector with eigenvalue 0.

3.3 The Associated Kernel

Define the subspace $\mathcal{S}_n := \{\mathbf{u} \in \mathbb{R}^n : \sum_{i=1}^n u_i = 0\}$.

Definition 3.5 (Laplacian Kernel). *The associated kernel is $K_G(i, j) = \mathbf{e}_i^\top L^+ \mathbf{e}_j = L_{ij}^+$, where L^+ denotes the Moore–Penrose pseudo-inverse of L .*

Proposition 3.6 (Laplacian Feature Map). *The map $\phi : [n] \rightarrow \mathcal{S}_n$ defined by $\phi(i) := \mathbf{e}_i^\top L^+$ is a feature map for the kernel K_G (Definition 3.5).*

Proof.

$$\langle \phi(i), \phi(j) \rangle_G = \langle \mathbf{e}_i^\top L^+, \mathbf{e}_j^\top L^+ \rangle_G = \mathbf{e}_i^\top L^+ L (\mathbf{e}_j^\top L^+)^\top = \mathbf{e}_i^\top L^+ L L^+ \mathbf{e}_j = \mathbf{e}_i^\top L^+ \mathbf{e}_j = L_{ij}^+$$

□

4 Part IV: Spectral Clustering*

4.1 Objectives

We aim to partition the graph into disjoint sets A and B .

Definition 4.1 (Graph Cut Objectives*). • **Mincut:** $\text{cut}(A, B) = \sum_{i \in A, j \in B} w_{ij}$.

• **RatioCut:** $\text{RatioCut}(A, B) = \text{cut}(A, B) \left(\frac{1}{|A|} + \frac{1}{|B|} \right)$.

Remark 4.2 (Computational Complexity*). *Mincut can be computed efficiently in $O(n^3)$ time, but tends to produce unbalanced clusters. Finding the minimum RatioCut is NP-hard, but captures balanced cluster structure.*

4.2 Relaxation of Balanced Cuts

We first reformulate the perfectly balanced cut problem. Represent a clustering by a vector $\mathbf{u} \in \{-1, 1\}^n$ with $u_i = 1$ if $i \in A$ and $u_i = -1$ if $i \in B$. Then by Theorem 3.3:

$$\text{cut}(A, B) = \frac{1}{4} \sum_{i < j} w_{ij} (u_i - u_j)^2 = \frac{1}{4} \mathbf{u}^\top L \mathbf{u} \quad (4)$$

and $|A| = |B|$ implies $\mathbf{u} \perp \mathbf{1}$.

Relaxing $\mathbf{u}$ to real values yields the efficiently solvable problem:

$$\min_{\mathbf{u} \in \mathbb{R}^n} \mathbf{u}^\top L \mathbf{u} \quad \text{s.t.} \quad \mathbf{u} \perp \mathbf{1}, \|\mathbf{u}\|^2 = n \quad (5)$$

Theorem 4.3 (Generalised Rayleigh–Ritz*). *If $\lambda_1 \leq \dots \leq \lambda_n$ are the eigenvalues of a real-valued symmetric matrix M with corresponding eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$, then:*

$$\lambda_k = \min_{\mathbf{u} \neq \mathbf{0}} \frac{\mathbf{u}^\top M \mathbf{u}}{\mathbf{u}^\top \mathbf{u}} \quad \text{subject to} \quad \mathbf{u} \perp \mathbf{v}_1, \dots, \mathbf{v}_{k-1} \quad (6)$$

Since $\mathbf{1}$ is the first eigenvector of the Laplacian (Proposition 3.4), by Theorem 4.3 the solution to (5) is the **second eigenvector** of L (the Fiedler vector), scaled by $\sqrt{n}$.

Remark 4.4 (Relaxed RatioCut*). *If we instead represent a clustering by $u_i = \sqrt{|B|/|A|}$ for $i \in A$ and $u_i = -\sqrt{|A|/|B|}$ for $i \in B$, then $\mathbf{u}^\top L \mathbf{u} = (|A| + |B|) \text{RatioCut}(A, B)$ and $\mathbf{u} \perp \mathbf{1}$. Therefore, (5) also provides a relaxed solution to RatioCut.*

In practice, the real-valued solution is converted to a discrete partition using $\text{sign}(\mathbf{u})$ or by finding a natural gap in the sorted values u_i .

5 Part V: Minimum Cut Transduction*

Definition 5.1 (Minimum Cut Classifier*). *Given labeled vertices $\{(v_{i_1}, y_1), \dots, (v_{i_\ell}, y_\ell)\}$ with $y \in \{-1, +1\}$, the hypothesis vector is:*

$$\hat{\mathbf{u}} = \arg \min_{\mathbf{u} \in \{-1, 1\}^n} \left\{ \sum_{(i,j) \in E(G)} w_{ij} |u_i - u_j| : u_{i_1} = y_1, \dots, u_{i_\ell} = y_\ell \right\} \quad (7)$$

This may be solved by a min-cut/max-flow algorithm in $O(n^3)$ time. Introduced by Blum and Chawla (2001).

- Remark 5.2** (Observations on Minimum Cut). 1. The minimum of the objective (7) is unchanged when relaxed from $\mathbf{u} \in \{-1, 1\}^n$ to $\mathbf{u} \in \mathbb{R}^n$.
2. The problem is ill-posed (no unique solution). For example, on a path graph with v_2 labeled $+1$ and v_5 labeled -1 , the value at v_3 is ambiguous.
3. Like the mincut objective for clustering (Remark 4.2), this method tends to produce unbalanced labelings.

6 Part VI: Laplacian-based Transduction*

6.1 Harmonic Energy Minimization*

To address the issues of the minimum cut approach (Remark 5.2), we replace the ℓ_1 penalty with the squared ℓ_2 penalty (Laplacian energy from Theorem 3.3).

Definition 6.1 (Harmonic Solution*). The interpolant vector is:

$$\bar{\mathbf{u}} = \arg \min_{\mathbf{u} \in \mathbb{R}^n} \left\{ \sum_{(i,j) \in E(G)} w_{ij}(u_i - u_j)^2 : u_{i_k} = y_k \text{ for } k = 1, \dots, \ell \right\} \quad (8)$$

Prediction at vertex v is $\hat{y}_v := \text{sign}(\bar{u}_v)$.

Theorem 6.2 (Harmonic Property*). The solution $\bar{\mathbf{u}}$ of Definition 6.1 satisfies:

$$\bar{u}_i = \frac{\sum_{j=1}^n w_{ij} \bar{u}_j}{d_i} \quad \text{for each unlabeled vertex } i = \ell + 1, \dots, n \quad (9)$$

and $\bar{u}_i = y_i$ for each labeled vertex $i = 1, \dots, \ell$.

Remark 6.3 (Names for this Method*). This method is known by many names, including harmonic minimization, label propagation, and Laplacian interpolated regularization. Variations on it account for a large portion of graph-based SSL; see, e.g., [2] and [6].

6.2 Consensus Algorithm (Label Propagation)*

The harmonic solution (Definition 6.1) can be computed via a simple iterative procedure.

Algorithm 1 Consensus Algorithm (Relaxation)

Initialize $u_i = y_i$ for labeled vertices; $u_i = 0$ for unlabeled vertices.

repeat

 Select a free (unlabeled) vertex i uniformly at random.

 Update: $u'(i) \leftarrow \frac{\sum_{j \in \Gamma(i)} w_{ij} u_j}{d_i}$

until convergence

Proposition 6.4 (Consensus Convergence*). Algorithm 1 converges to the harmonic solution $\bar{\mathbf{u}}$ (Definition 6.1).

Convergence Sketch. Define the “energy” $E(\mathbf{u}) = \|\mathbf{u}\|_G^2$ (Definition 3.2). Let $\mathbf{u}, \mathbf{u}', \mathbf{u}''$ be successive labelings. Each averaging step reduces the energy: $\|\mathbf{u}\|_G^2 \geq \|\mathbf{u}'\|_G^2 \geq \|\mathbf{u}''\|_G^2 \geq \dots \geq 0$. Since the sequence is monotonically decreasing and bounded below by 0, it converges in energy. The limit is the unique minimizer $\bar{\mathbf{u}}$. $\square$

Exercise 6.5 (Four Node Problem). Consider the path graph: $+1 - a - b - -1$. Using the consensus algorithm (Algorithm 1), determine what values nodes a and b converge to, and justify your answer using the harmonic property (Theorem 6.2).

7 Part VII: Interpreting Laplacian-based Transduction*

Several equivalent perspectives explain why Laplacian-based transduction (Definition 6.1) works:

1. As a resistive network (Section 7.1)
2. As a random walk on a network (Section 7.3)
3. Under certain conditions, the graph-Laplacian approximates the Riemannian Laplacian of the data manifold
4. Via predictive performance guarantees (Section ??)

7.1 Resistive Networks*

Identify the graph with a resistive network.

- Edges are resistors with conductance w_{ij} (resistance $\pi_{ij} = 1/w_{ij}$).
- Labels are voltage constraints.
- Power of a labeling:

$$P(\mathbf{u}) := \|\mathbf{u}\|_G^2 = \sum_{(i,j) \in E(G)} w_{ij} |u_i - u_j|^2 = \mathbf{u}^\top L \mathbf{u} \quad (10)$$

Label by minimizing power subject to constraints, which is exactly the harmonic solution (Definition 6.1).

7.2 Effective Resistance*

Definition 7.1 (Effective Resistance*). *For a pair of vertices p and q , the effective resistance $r_G(p, q)$ between them is the resistance of the total network when a voltage source is connected across them; equivalently, it is the voltage difference needed to induce a unit current flow between p and q .*

The effective resistance can also be computed via optimization:

$$r_G(p, q) = \frac{1}{\min_{\mathbf{u} \in \mathbb{R}^n} \{ \|\mathbf{u}\|_G^2 : u_p = 1, u_q = 0 \}} \quad (11)$$

7.2.1 Connection to Kernel L^+

We now prove the fundamental relationship between effective resistance (Definition 7.1) and the Laplacian pseudo-inverse (Definition 3.5).

Lemma 7.2 (Optimization Lemma). *Let $x \in \mathcal{H}$ (a Hilbert space), $x \neq 0$. Then*

$$\|x\|^{-2} = \min_{w \in \mathcal{H}} \{ \|w\|^2 : \langle w, x \rangle = 1 \} \quad (12)$$

Proof. Observe that $w^* = \frac{x}{\|x\|^2}$ is a feasible solution since $\langle \frac{x}{\|x\|^2}, x \rangle = \frac{\|x\|^2}{\|x\|^2} = 1$. The norm is $\|w^*\|^2 = \frac{\|x\|^2}{\|x\|^4} = \|x\|^{-2}$.

Suppose $v = \frac{x}{\|x\|^2} + \alpha z$ is a feasible solution, where $\langle z, x \rangle = 0$ (to maintain feasibility $\langle v, x \rangle = 1$). Then:

$$\|v\|^2 = \left\| \frac{x}{\|x\|^2} + \alpha z \right\|^2 = \frac{\|x\|^2}{\|x\|^4} + 2\alpha \frac{\langle x, z \rangle}{\|x\|^2} + \alpha^2 \|z\|^2 = \|x\|^{-2} + \alpha^2 \|z\|^2$$

This is minimized when $\alpha = 0$ (or $z = 0$), confirming the minimum is $\|x\|^{-2}$. $\square$

Theorem 7.3 (Effective Resistance via Kernel, KR93*). *The effective resistance (Definition 7.1) between v_p and v_q is:*

$$r_G(p, q) = L_{pp}^+ + L_{qq}^+ - 2L_{pq}^+ \quad (13)$$

i.e., the effective resistance is the squared distance in the feature space induced by the kernel L^+ (Definition 3.5).

Proof of Theorem 7.3.* From (11),

$$r_G(p, q) = \frac{1}{\min_{\mathbf{u} \in \mathbb{R}^n} \{\|\mathbf{u}\|_G^2 : u_p = 1, u_q = 0\}}$$

Recall $\mathcal{S}_n = \{\mathbf{u} : \sum u_i = 0\}$. Since $\|\mathbf{u}\|_G^2 = \|\mathbf{u} + c\mathbf{1}\|_G^2$ for any constant c , the constraint $u_p = 1, u_q = 0$ is equivalent to $u_p - u_q = 1$, and we may restrict to $\mathbf{u} \in \mathcal{S}_n$:

$$\min_{\mathbf{u} \in \mathbb{R}^n} \{\|\mathbf{u}\|_G^2 : u_p = 1, u_q = 0\} = \min_{\mathbf{u} \in \mathcal{S}_n} \{\|\mathbf{u}\|_G^2 : u_p - u_q = 1\}$$

Using $\langle \mathbf{e}_i^\top L^+, \mathbf{u} \rangle_G = \mathbf{e}_i^\top L^+ L \mathbf{u} = u_i$ for any $\mathbf{u} \in \mathcal{S}_n$ (by Proposition 3.6):

$$u_p - u_q = \langle \mathbf{e}_p^\top L^+ - \mathbf{e}_q^\top L^+, \mathbf{u} \rangle_G$$

Let $x = \mathbf{e}_p^\top L^+ - \mathbf{e}_q^\top L^+$. The constraint becomes $\langle x, \mathbf{u} \rangle_G = 1$. By Lemma 7.2, the minimum value of the constrained problem is $\|x\|_G^{-2}$. Thus:

$$r_G(p, q) = \frac{1}{\|x\|_G^{-2}} = \|x\|_G^2$$

Expanding:

$$\begin{aligned} \|\mathbf{e}_p^\top L^+ - \mathbf{e}_q^\top L^+\|_G^2 &= (\mathbf{e}_p^\top L^+ - \mathbf{e}_q^\top L^+) L (\mathbf{e}_p^\top L^+ - \mathbf{e}_q^\top L^+)^\top \\ &= (\mathbf{e}_p^\top L^+ - \mathbf{e}_q^\top L^+) (\mathbf{e}_p - \mathbf{e}_q) \quad (\text{since } L^+ L L^+ = L^+) \\ &= L_{pp}^+ - L_{pq}^+ - L_{qp}^+ + L_{qq}^+ \\ &= L_{pp}^+ + L_{qq}^+ - 2L_{pq}^+ \end{aligned}$$

□

7.2.2 Graph Connectivity via Effective Resistance

The kernel L^+ encodes connectivity information about the graph.

Definition 7.4 (Inverse Connectivities). • **Total graph resistance:** $R_{\text{tot}} := \sum_{i>j} r_G(i, j)$

• **Vertex resistance:** $R(v) := \sum_{i=1}^n r_G(v, i)$

• **Relation:** $L_{vv}^+ = \frac{R(v)}{n} - \frac{R_{\text{tot}}}{n^2}$

7.3 Random Walks*

Definition 7.5 (Transition Probability). *The transition probability P_{ij} of going from vertex i to vertex j on a step of a random walk is:*

$$P_{ij} := \frac{w_{ij}}{\sum_k w_{ik}} = \frac{w_{ij}}{d_i} \quad (14)$$

where W is the symmetric weighted adjacency matrix.

Theorem 7.6 (Random Walk Interpretation*). *Given an unweighted graph G with adjacency matrix W , a set of vertices V_0 labeled “0”, and a set of vertices V_1 labeled “1”, let p_i denote the probability that a random walk started at vertex i reaches a vertex in V_1 before reaching a vertex in V_0 . Then $\mathbf{p}$ is the harmonic solution (Definition 6.1):*

$$\mathbf{p} = \arg \min_{\mathbf{p} \in \mathbb{R}^n} \left\{ \sum_{i < j} w_{ij} |p_i - p_j|^2 : (p_k = 0)_{v_k \in V_0}, (p_k = 1)_{v_k \in V_1} \right\} \quad (15)$$

Proof. The equations and boundary conditions determining the random walk are isomorphic to those of label propagation (Theorem 6.2). At any unlabeled vertex i , the random walk probability satisfies $p_i = \sum_j P_{ij} p_j = \frac{\sum_j w_{ij} p_j}{d_i}$, which is precisely the harmonic property (9). $\square$

Theorem 7.7 (Commute Time*). *The commute time $c_G(i, j)$ (expected number of steps to travel from i to j and back) satisfies:*

$$c_G(i, j) = \text{Vol}(G) \cdot r_G(i, j) \quad (16)$$

where $\text{Vol}(G) = \sum_{i,j} w_{ij}$ is twice the total degree and $r_G(i, j)$ is the effective resistance (Definition 7.1).

7.4 Intuition: Cluster Structure*

Effective resistance adapts the base distance metric $d(p, q)$ according to the empirical connectivity of the graph via $r_G(p, q)$.

Example 7.8 (Two Cliques*). *Consider two m -cliques connected by a single edge with distance parameters $a < b$. By the parallel resistance law, within a clique whose edge-resistance is $\leq d$, the effective resistance is $r_{m\text{-clique}} \leq \frac{2d}{m}$. If $\frac{2b}{m} < a$, then the bottleneck dominates and the node between the cliques receives label “+1”. Labelings thus respect cluster structure: the effective resistance “contracts” densely connected regions and “expands” bottlenecks.*

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